

A COMPLETE SET OF MUTUALLY UNBIASED BASES IN DIMENSION 27 WHOSE PROJECTIVE TORIC DESIGN IS NEITHER A GROUP NOR A COSET OF ONE

ABSTRACT. Iosue, Mooney, Ehrenberg and Gorshkov conjectured that a projective toric 2-design $X \subseteq \mathbb{P}(T^d)$ of size $|X| = d^2$ which is not a subgroup of $\mathbb{P}(T^d)$ cannot satisfy their angle condition. As printed the statement is trivially false, since the two conditions on X depend only on differences of points and so are insensitive to translation, while “subgroup” is not; a translated group design at $d = 2$ already refutes it. We refute the evident repair, in which “subgroup” is replaced by “coset of a subgroup”, at $d = 27$. The witness is the 729-point projective toric design of the complete set of 28 mutually unbiased bases in $\mathbb{C}^{27}$ built from the Ball–Bamberg–Lavrauw–Penttilä symplectic spread of the Hering plane of order 27: it contains the identity, is a 2-design of size 27^2 , satisfies the angle condition, and is neither a subgroup nor a coset of a subgroup. The obstruction is that the 27 linearised maps of that spread are not closed under addition, because -1 is a nonsquare in $\text{GF}(27)$ and $h_m + h_m = h_{-m}$. All data determining the 729 points are printed here; the nonclosure argument is by hand, while the design and angle conditions are exact finite checks, carried out by the accompanying program.

1. THE CONJECTURE

Let $T = \{z \in \mathbb{C} : |z| = 1\}$, let T^d be the d -torus written additively as angles, and let $\mathbb{P}(T^d) = T^d/T$ be the quotient by the diagonal subgroup, with representatives normalised so that the first angle is 0 [5, Definition 2.4]. For $k \in \mathbb{Z}^d$ put $\chi_k(\phi) = e^{ik \cdot \phi}$. A finite $X \subseteq \mathbb{P}(T^d)$ is a *projective toric 2-design* if

$$(1) \quad \sum_{\phi \in X} \chi_k(\phi) = 0 \quad \text{for every nonzero } k = e_{a_1} + e_{a_2} - e_{b_1} - e_{b_2},$$

the indices a_1, a_2, b_1, b_2 ranging over $\{1, \dots, d\}$ [5, Definition 2.6]. Write (2) for the angle condition of [5, Theorem 4.4],

$$(2) \quad \forall \phi \neq \theta \in X: \quad \left| \sum_{j=1}^d e^{i(\phi_j - \theta_j)} \right|^2 \in \{0, d\}.$$

Theorem 4.4 of [5] says that a complete set of $d+1$ mutually unbiased bases (MUBs) in $\mathbb{C}^d$ exists if and only if some $X \subseteq \mathbb{P}(T^d)$ with $|X| = d^2$ satisfies

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both (1) and (2). The statement at issue is the following, quoted verbatim from [5].

Conjecture 4.9. Let d be any dimension and let X be a $\mathbb{P}(T^d)$ 2-design of size $|X| = d^2$. If X is not a subgroup of $\mathbb{P}(T^d)$, then X does not satisfy (2).

The lead-in in [5] reads “we conjecture that, in any dimension, a complete set of MUBs must come from a group projective toric 2-design”, and the intended payoff is recorded there: with [5, Proposition 4.6] the conjecture would prove that no complete set of MUBs exists in dimension 6. It is quoted as open in the survey [8].

Two failures must be separated. Conditions (1) and (2) depend only on differences of points, so they are invariant under translation of X , whereas being a subgroup is not; hence any translate of a group design that misses the identity refutes the conjecture as printed, and $X_2 = \{\pi/4, 3\pi/4, 5\pi/4, 7\pi/4\}$ in $\mathbb{P}(T^2)$ (second angle; the first is 0) is such a translate. That is a normalisation slip, repaired for free by replacing “subgroup” by “coset of a subgroup”: X_2 is such a coset. The witness below is what settles the repaired statement, since it contains the identity and fails anyway.

Theorem 1.1. *Conjecture 4.9 of [5] is false, and remains false with “subgroup” replaced by “coset of a subgroup”. Explicitly, let X be the 729-point subset of $\mathbb{P}(T^{27})$ determined by the data of Section 2. Then $|X| = 27^2$, X satisfies (1) and (2), X contains the identity of $\mathbb{P}(T^{27})$, and X is neither a subgroup of $\mathbb{P}(T^{27})$ nor a coset of one. By [5, Theorem 4.4], X is the projective toric design of a complete set of 28 mutually unbiased bases in $\mathbb{C}^{27}$.*

2. THE WITNESS

Every angle occurring below is an integer multiple of $2\pi/3$, so a point of $\mathbb{P}(T^{27})$ is recorded as a word of 27 digits in $\mathbb{F}_3 = \{0, 1, 2\}$, the j -th digit being the j -th angle in units of $2\pi/3$.

Let $\mathbb{F} = \text{GF}(27) = \mathbb{F}_3[t]/(t^3 - t - 1)$; the cubic is irreducible because it sends 0, 1, 2 to 2, 2, 2. Index the elements of $\mathbb{F}$ by

$$(3) \quad e = a_0 + 3a_1 + 9a_2 \longleftrightarrow a_0 + a_1t + a_2t^2, \quad a_i \in \{0, 1, 2\},$$

and use this order for the 27 coordinates of $\mathbb{P}(T^{27})$, so that coordinate $w = 0$ is the normalised one. Let $\text{tr}(x) = x + x^3 + x^9$ be the absolute trace.

Table 1 lists 27 maps $Q: \mathbb{F} \rightarrow \mathbb{F}_3$, one per row, as the digit word $(Q(w))_w$ in the order (3). They are

$$(4) \quad \begin{aligned} Q = \{0\} \cup \{ w \mapsto \text{tr}(m^3w^4) : m \text{ a nonsquare of } \mathbb{F} \} \\ \cup \{ w \mapsto \text{tr}(2ws\beta(ws)) : s \in \mathbb{F}^*/\{\pm 1\} \}, \end{aligned}$$

where $\beta(v) = \frac{1}{2}(-v + v^3 + v^9) = 2(-v + v^3 + v^9)$, the row labels recording m and s ; there are $1 + 13 + 13 = 27$ of them.

TABLE 1. The 27 quadratic forms Q of (4), each as the digit word $(Q(w))_{w \in \text{GF}(27)}$ in the index order (3).

Q0	00000000000000000000000000000000	Qs1	000222222012201201021210210
Qm2	000111111210021021201012012	Qs3	022022022220001112202121010
Qm6	000210201201111102210120111	Qs4	022202220010112022001022121
Qm7	000102120021120111012111102	Qs5	022220202121022001112010022
Qm8	000021012111102120111102120	Qs9	000201210222012210222201021
Qm10	022112121112010100121100001	Qs10	011212221221200020212002200
Qm14	022121112001211010010001211	Qs11	022001010211220202211220202
Qm17	022211211100100211100211100	Qs12	022010001022121220022202112
Qm18	011110101011002212011221020	Qs13	000012021102222012120021222
Qm19	011002020101011221110212011	Qs14	011122122200122200200200122
Qm21	011200200110221110101101212	Qs15	022100100202202121220112220
Qm23	011011011020212101002110221	Qs16	000120102120210222102222201
Qm24	011020002212110011221011101	Qs17	011221212002020122020122002
Qm26	011101110122101002122020110		

Let $\Lambda = \{w \mapsto \text{tr}(vw) : v \in \mathbb{F}\}$ be the group of all 27 $\mathbb{F}_3$ -linear functionals on $\mathbb{F}$. As digit words it is the $\mathbb{F}_3$ -span of the three rows

$$\begin{aligned} v = 1 & \quad 00000000022222222111111111 \\ v = t & \quad 000222111000222111000222111 \\ v = t^2 & \quad 021021021210210210102102102 \end{aligned}$$

and $|\Lambda| = 27$. Finally set

$$(5) \quad X = \{Q + \lambda : Q \in \mathcal{Q}, \lambda \in \Lambda\} \subseteq \mathbb{P}(T^{27}),$$

sums taken digitwise modulo 3. Table 1 and the three rows above determine X completely.

Proposition 2.1. $|X| = 729 = 27^2$; every point of X has $w = 0$ coordinate 0; the all-zero word, the identity of $\mathbb{P}(T^{27})$, lies in X ; X satisfies (2), the quantity in (2) taking the value 0 on 9 477 and the value 27 on 255 879 of the $\binom{729}{2} = 265\,356$ unordered pairs and no other value; and X satisfies (1), all 142 506 nonzero characters $k = e_{a_1} + e_{a_2} - e_{b_1} - e_{b_2}$ summing to zero over X .

Proof. These are exhaustive finite verifications on the data of Table 1 and the three generators of Λ , each an exact count in $\mathbb{Z}$ rather than a numerical evaluation. Writing $\omega = e^{2\pi i/3}$ and, for integers n_0, n_1, n_2 ,

$$\begin{aligned} n_0 + n_1\omega + n_2\omega^2 = 0 & \iff n_0 = n_1 = n_2, \\ |n_0 + n_1\omega + n_2\omega^2|^2 &= \sum_i n_i^2 - \sum_{i < j} n_i n_j, \end{aligned}$$

each sum in (1) and (2) is decided by the three residue counts of the relevant digit vector. For (2) let (n_0, n_1, n_2) count the residues of $\phi_w - \theta_w$ over the 27 coordinates, so $\sum n_i = 27$; over all 265 356 pairs exactly two shapes occur, namely (9, 9, 9) on 9 477 pairs, where the value is 0, and the permutations of (6, 9, 12) on the remaining 255 879 pairs, where the value is

$144 + 81 + 36 - 108 - 54 - 72 = 27$. There are no other shapes and hence no violations. For (1), the 27 coordinates give $\binom{28}{2} = 378$ size-2 multisets and hence $378^2 = 142884$ characters, of which exactly 378 have $k = 0$ — and those are precisely the 378 with $k \equiv 0 \pmod{3}$, so “nonzero” is unambiguous; each of the remaining 142506 has residue counts exactly $(243, 243, 243)$ over the 729 points, so its character sum is $243(1 + \omega + \omega^2) = 0$. The identity is the row Q_0 plus $\lambda = 0$. Every row of Table 1 and every generator of Λ begins with the digit 0, hence so does every point of X . The program of Section 5 carries out these counts. $\square$

3. WHY X IS NOT A GROUP

For an $\mathbb{F}_3$ -linear map $h: \mathbb{F} \rightarrow \mathbb{F}$ put $Q_h(w) = \text{tr}(\frac{1}{2} w h(w)) = \text{tr}(2 w h(w))$, and let

$$(6) \quad H = \{0\} \cup \{h_m : m \text{ a nonsquare}\} \cup \{h_s : s \in \mathbb{F}^* / \{\pm 1\}\},$$

$$h_m(x) = mx^9 + m^3x^3, \quad h_s(x) = s\beta(xs),$$

so that $|H| = 27$ and $\mathcal{Q} = \{Q_h : h \in H\}$, row by row. Call an $\mathbb{F}_3$ -linear map h *self-adjoint* if $\text{tr}(w h(w')) = \text{tr}(w' h(w))$ for all $w, w' \in \mathbb{F}$, and let S be the set of self-adjoint maps. That defining identity is $\mathbb{F}_3$ -linear in h , so S is an $\mathbb{F}_3$ -subspace of $\text{End}_{\mathbb{F}_3}(\mathbb{F})$; the program of Section 5 checks that the 27 maps (6) lie in S , so S contains H together with its additive span. Items (ii) and (iii) of Lemma 3.1 are asserted only for maps in S . The set H is the spread set of the Ball–Bamberg–Lavrauw–Penttila symplectic spread of $\mathbb{F} \times \mathbb{F}$, i.e. of the Hering plane of order 27; its 28 components are $\{(0, y)\}$, $\{(x, 0)\}$ and the graphs of the 26 nonzero maps in (6), as printed in [1, Corollary 1] after [2].

Lemma 3.1. *With the notation above:*

- (i) $h \mapsto Q_h$ is $\mathbb{F}_3$ -linear;
- (ii) for $h \in S$, the polar form of Q_h is $B_h(w, w') = Q_h(w + w') - Q_h(w) - Q_h(w') = \text{tr}(w h(w'))$;
- (iii) for $h, h' \in S$, $Q_h - Q_{h'} \in \Lambda$ if and only if $h = h'$;
- (iv) consequently the set X of (5) is a subgroup of $\mathbb{P}(T^{27})$ if and only if H is closed under addition;
- (v) since $0 \in X$, X is a coset of a subgroup if and only if it is a subgroup.

Step (ii) uses that $\text{char } \mathbb{F}$ is odd.

Proof. (i) is immediate. For (ii), expand $Q_h(w + w') - Q_h(w) - Q_h(w') = \text{tr}(2(w h(w') + w' h(w)))$ and use the hypothesis $h \in S$, that is $\text{tr}(w h(w')) = \text{tr}(w' h(w))$; the right-hand side is $\text{tr}(4 w h(w')) = \text{tr}(w h(w'))$ because $4 \equiv 1 \pmod{3}$. Here $\frac{1}{2}$ must exist in $\mathbb{F}_3$, which is where odd characteristic enters. For (iii), let $h, h' \in S$; if $Q_h - Q_{h'} = \lambda \in \Lambda$ then its polar form vanishes, so by (ii) $\text{tr}(w(h - h')(w')) = 0$ for all w, w' , and nondegeneracy of tr gives $h = h'$; conversely $Q_h - Q_h = 0 \in \Lambda$. For (iv): Λ is a subgroup, and by (iii), applied inside $S \supseteq H$, the 27 cosets $Q_h + \Lambda$ are pairwise disjoint, so

$X = \bigsqcup_{h \in H} (Q_h + \Lambda)$ has 729 elements and is closed under addition if and only if for all $h, h' \in H$ there is $h'' \in H$ with $Q_h + Q_{h'} - Q_{h''} \in \Lambda$; since $h + h'$ and h'' again lie in S , (i) and (iii) turn this into exactly $h + h' \in H$. For (v): if $X = x_0 + G$ with G a subgroup, then $0 \in X$ gives $-x_0 \in G$, hence $x_0 \in G$ and $X = x_0 + G = G$. $\square$

Remark 3.2 (attribution). Lemma 3.1 is not new: it is the classical dictionary between abelian semiregular relative $(d, d, d, 1)$ -difference sets, planar functions and commutative semifields, in the form used for MUBs by Godsil and Roy [4, §6] and by Klappenecker and Rötteler [7], going back to [3], and it must be cited together with Kantor’s published correction to [4, p. 255], which restricts that correspondence to *semifield* spreads [6]. Kantor [6] gives the exponent parameterisation of the MUBs of a symplectic spread, which is $X = \{Q_h + \lambda\}$ written in exponents; states the spread-side half of (iv)–(v), that a symplectic spread yields a group-like design exactly when it is a semifield spread; and names the BBLP family among spreads “not related to semifields”. Abdukhalikov [1] prints the witness spread and its non-semifield property. Neither paper mentions projective toric designs, neither states the subgroup or coset condition on $\mathbb{P}(T^d)$, and neither draws the conclusion of Theorem 1.1. Beyond that conclusion, nothing here is new.

Three facts about $\text{GF}(27)$ finish the proof.

Proof of Theorem 1.1. By Proposition 2.1, X is a $\mathbb{P}(T^{27})$ 2-design of size 27^2 which satisfies (2) and contains the identity. It remains to show that H is not closed under addition; Lemma 3.1(iv),(v) then says that X is neither a subgroup nor a coset of one, and Conjecture 4.9 fails at $d = 27$.

Fact 1: -1 is a nonsquare in $\text{GF}(27)$. The group $\mathbb{F}^*$ is cyclic of order 26, so the squares form its subgroup of index 2 and *odd* order 13, while -1 has order 2; as $2 \nmid 13$, -1 is a nonsquare. (In the labelling (3), -1 is the element 2.)

Fact 2: $h_m + h_m = h_{-m}$ for every $m \in \mathbb{F}$. In characteristic 3, $h_m + h_m = 2mx^9 + 2m^3x^3 = (2m)x^9 + (2m)^3x^3 = h_{2m} = h_{-m}$, using $(2m)^3 = 8m^3 = 2m^3$.

Fact 3: the two families in (6) are separated by one coefficient. Every $\mathbb{F}_3$ -linear map $\mathbb{F} \rightarrow \mathbb{F}$ has a unique linearised representation $a_0x + a_1x^3 + a_2x^9$: there are $27^3 = 3^9$ such expressions, all $\mathbb{F}_3$ -linear and pairwise distinct, and $|\text{End}_{\mathbb{F}_3}(\text{GF}(27))| = 3^9$. In that representation every h_m has $a_0 = 0$ (and $a_1 = m^3$, $a_2 = m$), while every h_s has $a_0 = s^2 \neq 0$.

Now take any nonsquare m ; by Fact 1 there are 13 of them, and $m = -1$ is one. Then $h_m \in H$ and, by Fact 2, $h_m + h_m = h_{-m}$ with $-m$ a square by Fact 1. Hence $h_{-m} \neq 0$; $h_{-m} \neq h_{m'}$ for any nonsquare m' , because $m' \mapsto h_{m'}$ is injective (read off $a_2 = m'$); and $h_{-m} \neq h_s$ for any s , by Fact 3. So $h_m + h_m \notin H$ and H is not closed under addition. $\square$

Table 1 exhibits the failing sum directly. Doubling the row **Qm2** digitwise modulo 3,

$$\begin{array}{rcl} 000111111210021021201012012 & + & 000111111210021021201012012 \\ & & 000222222120012012102021021, \end{array} =$$

produces the form $Q_{h_{-m}}$ for $m = 2 = -1$, which is not a row of Table 1; since $h_{-m} = h_m + h_m$ lies in S but not in H , Lemma 3.1(iii) puts it in no coset $Q_h + \Lambda$ with $h \in H$, so it lies outside X .

4. SCOPE

Nothing here bears on the existence of a complete set of MUBs in dimension 6; what is removed is one route to a negative answer. Conjecture 4.7 of [5], quantified over non-prime-power d , is a different statement and is untouched. Characteristic 2, and with it $d = 8$, is not attempted: Lemma 3.1(ii) needs odd characteristic. We do not claim that no complete set of MUBs MUB-equivalent to the one behind X has a subgroup design; the statement is about the specific X of (5), and the MUB-equivalence group is a continuum that was not searched. We do not claim that $d = 27$ is least; no other dimension is examined here.

5. VERIFICATION

The accompanying program `verify.py` (Python 3.9+, standard library only, no external data file) takes as input only what is printed above: the presentation $\mathbb{F}_3[t]/(t^3-t-1)$ with the index (3), the 27 digit words of Table 1, and the three generators of Λ . From that data it rebuilds $\text{GF}(27)$, its trace and its squares; checks that the printed rows are exactly the maps (4) and are reproduced by $Q_h(w) = \text{tr}(2wh(w))$ from (6); checks that each row is a quadratic form whose polar form is symmetric and equals $\text{tr}(wh(w'))$, so that each map (6) lies in S , and that the 27 polar forms give 27 distinct symmetric 3×3 matrices over $\mathbb{F}_3$ whose $\binom{27}{2} = 351$ pairwise differences are all nonsingular, so that (6) really is a symplectic spread; builds X and confirms Proposition 2.1 by the exhaustive sweeps quoted there; confirms Facts 1–3 and that X is not additively closed, which with $0 \in X$ and Lemma 3.1(v) also settles the coset claim; and confirms the $d = 2$ example of Section 1 in $\mathbb{Z}[\zeta_8]$. All arithmetic is exact in $\mathbb{Z}$, $\mathbb{F}_3$ or $\mathbb{Z}[\zeta_8]$; no floating-point value decides anything. The recorded run in `verify.output.txt` reports **VERDICT: ALL 35 CHECKS PASS** and closes with its own statement of what it does not cover.

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